STAT 4080/5080, SPRING 2026 **NAME**_____

FINAL EXAM **SIGNATURE**_____

INSTRUCTIONS

1. You are not allowed to communicate in any way with anyone in or outside of this class. Anyone found discussing the exam with other people is considered cheating, and thus all parties involved in the discussion will receive a "0" score on the exam. And all parties found violating the codes of academic honesty will be reported to respective colleges.

2. You should type the answers directly onto a word document. The answers have to be as specific as possible. All the relevant outputs should be copied and pasted onto the same document, with clear indications as to where to find those outputs. You have to show all the work, including writing down the hypotheses, in order to earn full credit. However, please only include relevant outputs. The total score is 92 for STAT 4060 and 100 for STAT 5060.

3. You can choose to use any software packages.

4. The final exam is by **2:00 p.m., Wednesday, April 29, 2026.** Please submit a hard copy of your completed final exam to me in person before the deadline. My office is located at Maurer Center 241C.

Question 1

We have a three-factor model where **Factor A** (random, a levels) and **Factor B** (random, b levels) are random, and **Factor C** (fixed, c levels) is fixed. The model is:

$$y_{ijkl} = \mu + \alpha_i + \beta_j + \gamma_k + (\alpha\beta)_{ij} + (\alpha\gamma)_{ik} + (\beta\gamma)_{jk} + (\alpha\beta\gamma)_{ijk} + \varepsilon_{(ijk)l}$$

where $i = 1, \ldots, a$; $j = 1, \ldots, b$; $k = 1, \ldots, c$; $l = 1, \ldots, n$, and the random effects $\alpha_i \sim N(0, \sigma^2_\alpha)$, $\beta_j \sim N(0, \sigma^2_\beta)$, $(\alpha\beta)_{ij} \sim N(0, \sigma^2_{\alpha\beta})$, $(\alpha\gamma)_{ik} \sim N(0, \sigma^2_{\alpha\gamma})$, $(\beta\gamma)_{jk} \sim N(0, \sigma^2_{\beta\gamma})$, $(\alpha\beta\gamma)_{ijk} \sim N(0, \sigma^2_{\alpha\beta\gamma})$, and $\varepsilon_{(ijk)l} \sim N(0, \sigma^2)$ are all independent.

Part (a): Complete the Tables

ANOVA Source Table (d.f. and SS)

Source of Variation	d.f.	SS
A	$a-1$	$bcn\sum_i(\bar{y}_{i\cdots} - \bar{y}_{\cdots})^2$
B	$b-1$	$acn\sum_j(\bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots})^2$
C	$c-1$	$abn\sum_k(\bar{y}_{\cdot\cdot k} - \bar{y}_{\cdots})^2$
A×B	$(a-1)(b-1)$	$cn\sum_i\sum_j(\bar{y}_{ij\cdot} - \bar{y}_{i\cdots} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots})^2$
A×C	$(a-1)(c-1)$	$bn\sum_i\sum_k(\bar{y}_{i\cdot k} - \bar{y}_{i\cdots} - \bar{y}_{\cdot\cdot k} + \bar{y}_{\cdots})^2$
B×C	$(b-1)(c-1)$	$an\sum_j\sum_k(\bar{y}_{\cdot jk} - \bar{y}_{\cdot j\cdot} - \bar{y}_{\cdot\cdot k} + \bar{y}_{\cdots})^2$
A×B×C	$(a-1)(b-1)(c-1)$	$n\sum_i\sum_j\sum_k(\bar{y}_{ijk\cdot} - \bar{y}_{ij\cdot} - \bar{y}_{i\cdot k} - \bar{y}_{\cdot jk} + \bar{y}_{i\cdots} + \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdot\cdot k} - \bar{y}_{\cdots})^2$
Error	$abc(n-1)$	$\sum\sum\sum\sum(y_{ijkl} - \bar{y}_{ijk\cdot})^2$
Total	$abcn-1$	$\sum\sum\sum\sum(y_{ijkl} - \bar{y}_{\cdots})^2$

EMS Column Table (Unrestricted model: A, B random; C fixed)

For the **unrestricted model**, the E(MS) includes σ^2 plus the variance component or fixed-effect term for the row itself, plus all terms that contain the row's effect and involve **at least one random factor**.

Since A and B are random and C is fixed, any interaction involving A or B is random. Only $(\alpha\beta\gamma)_{ijk}$ is random because it involves A and B even though C is fixed.

Factor	R a i	R b j	F c k	R n l	E(MS)
α_i	1	b	c	n	$\sigma^2 + n\sigma^2_{\alpha\beta\gamma} + cn\sigma^2_{\alpha\beta} + bn\sigma^2_{\alpha\gamma} + bcn\sigma^2_\alpha$
β_j	a	1	c	n	$\sigma^2 + n\sigma^2_{\alpha\beta\gamma} + cn\sigma^2_{\alpha\beta} + an\sigma^2_{\beta\gamma} + acn\sigma^2_\beta$
γ_k	a	b	0	n	$\sigma^2 + n\sigma^2_{\alpha\beta\gamma} + an\sigma^2_{\beta\gamma} + bn\sigma^2_{\alpha\gamma} + Q_\gamma$
$(\alpha\beta)_{ij}$	1	1	c	n	$\sigma^2 + n\sigma^2_{\alpha\beta\gamma} + cn\sigma^2_{\alpha\beta}$
$(\alpha\gamma)_{ik}$	1	b	0	n	$\sigma^2 + n\sigma^2_{\alpha\beta\gamma} + bn\sigma^2_{\alpha\gamma}$
$(\beta\gamma)_{jk}$	a	1	0	n	$\sigma^2 + n\sigma^2_{\alpha\beta\gamma} + an\sigma^2_{\beta\gamma}$
$(\alpha\beta\gamma)_{ijk}$	1	1	0	n	$\sigma^2 + n\sigma^2_{\alpha\beta\gamma}$
$\varepsilon_{(ijk)l}$	1	1	1	1	σ^2

where $Q_\gamma = \dfrac{abn\sum_{k=1}^{c}\gamma_k^2}{c-1}$.

Part (b): Do Exact Tests Exist?

No, exact tests do not exist for any of the main effects (A, B, or C) in this specific unrestricted model.

Factor A: $E(\text{MS}_A) = \sigma^2 + n\sigma^2_{\alpha\beta\gamma} + cn\sigma^2_{\alpha\beta} + bn\sigma^2_{\alpha\gamma} + bcn\sigma^2_{\alpha}$. Under $H_0 : \sigma^2_\alpha = 0$, we need a denominator whose E(MS) equals $\sigma^2 + n\sigma^2_{\alpha\beta\gamma} + cn\sigma^2_{\alpha\beta} + bn\sigma^2_{\alpha\gamma}$. No single mean square in the table has this form (it would require both $\sigma^2_{\alpha\beta}$ and $\sigma^2_{\alpha\gamma}$ terms together without $\sigma^2_{\alpha\beta\gamma}$ being separable). Therefore, **no exact test exists for Factor A**; a Satterthwaite approximate F test is needed: $\text{MS}_{AB} + \text{MS}_{AC} - \text{MS}_{ABC}$ serves as the approximate denominator.

Factor B: By symmetry with Factor A (both are random with the same structure), **no exact test exists for Factor B**. The approximate denominator is $\text{MS}_{AB} + \text{MS}_{BC} - \text{MS}_{ABC}$.

Factor C (fixed): $E(\text{MS}_C) = \sigma^2 + n\sigma^2_{\alpha\beta\gamma} + an\sigma^2_{\beta\gamma} + bn\sigma^2_{\alpha\gamma} + Q_\gamma$. Under $H_0 : \gamma_k = 0$, the residual E(MS) is $\sigma^2 + n\sigma^2_{\alpha\beta\gamma} + an\sigma^2_{\beta\gamma} + bn\sigma^2_{\alpha\gamma}$. No single MS has exactly this form. Therefore, **no exact test exists for Factor C** either; the Satterthwaite approximation using $\text{MS}_{AC} + \text{MS}_{BC} - \text{MS}_{ABC}$ is used as the approximate denominator.

Question 2

We study light output of oscilloscope tubes. **Glass Type** (A, $a = 3$ levels, fixed) and **Temperature** (B, $b = 3$ levels: 100, 125, 150°C, fixed), with $n = 3$ replicates per cell. The two-way complete (interaction) model is:

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk}, \quad i = 1, 2, 3,\ j = 1, 2, 3,\ k = 1, 2, 3$$

where $\varepsilon_{ijk} \overset{iid}{\sim} N(0, \sigma^2)$.

```
## Cell means (Glass Type x Temperature):
##        100      125       150
## 1 572.6667 1087.333 1386.0000
## 2 553.0000 1035.000 1313.0000
## 3 573.3333 1054.667  886.6667
```

Part (a): Interaction Plot and Model Decision

Interpretation: The lines for Glass Types 1 and 2 are roughly parallel across temperature, both showing increasing light output. However, the line for Glass Type 3 diverges sharply — it rises from 100°C to 125°C but drops dramatically at 150°C compared to Types 1 and 2. The non-parallel, crossing pattern indicates a likely interaction between Glass Type and Temperature. We therefore proceed with the **two-way complete model** (including the interaction term).

Part (b): Test for Lack of Fit

Hypotheses:

$$H_{01} : y_{ijk} = \mu + \varepsilon_{ijk} \quad \text{vs.} \quad H_{a1} : y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk}$$

The test statistic is:

$$F_{01} = \frac{(\text{SSA} + \text{SSB} + \text{SSAB})/(ab-1)}{\text{MSE}} \sim F_{ab-1,\, ab(n-1)} \text{ under } H_{01}$$

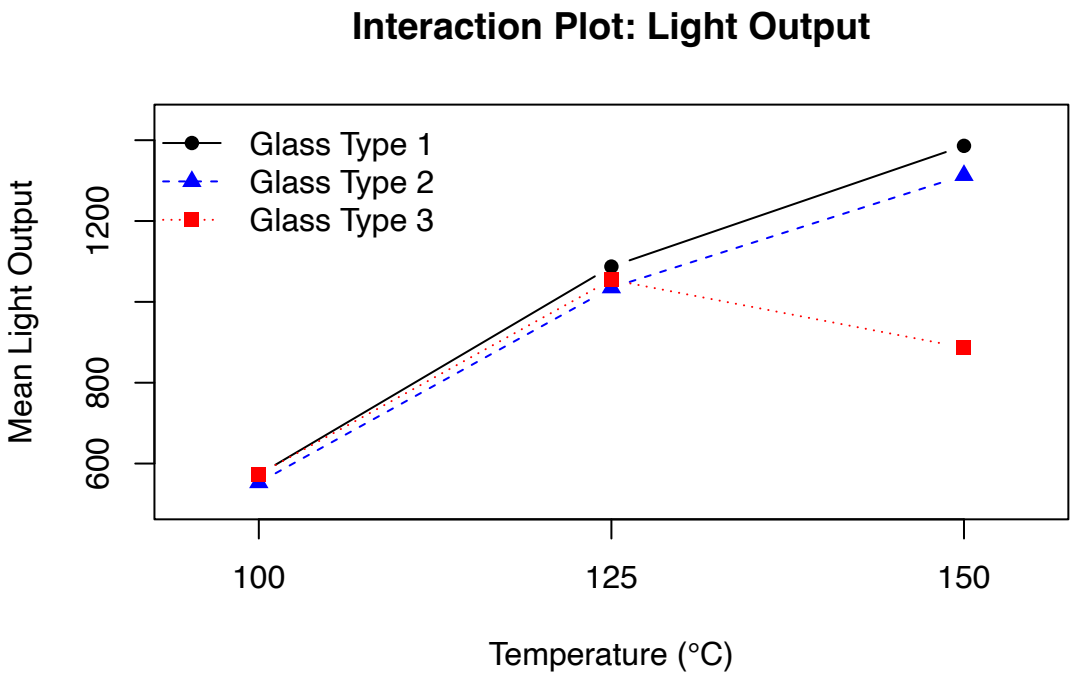


Figure 1: Interaction Plot: Light Output by Glass Type and Temperature

```
## Two-Way Complete Model ANOVA Table:
##                  Df  Sum Sq Mean Sq F value    Pr(>F)
## glass_type        2  150865   75432  206.37 3.886e-13 ***
## temp              2 1970335  985167 2695.26 < 2.2e-16 ***
## glass_type:temp   4  290552   72638  198.73 1.254e-14 ***
## Residuals        18    6579     366
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Lack-of-Fit Test:
##   F = 824.7704,  df = (8, 18),  p-value = 1.7833e-21
```

Conclusion: $F_{01} = 824.77$ with degrees of freedom $(8, 18)$, $p < 0.0001$. We strongly reject H_{01}. The two-way complete model provides a significantly better fit than the null (intercept-only) model. The model is significant and we proceed with the complete model.

Part (c): Test for Interaction

Hypotheses:

$$H_{02} : (\alpha\beta)_{ij} - \overline{(\alpha\beta)}_{i.} - \overline{(\alpha\beta)}_{.j} + \overline{(\alpha\beta)}_{..} = 0 \text{ for all } i, j \quad \text{vs.} \quad H_{a2} : \text{not } H_{02}$$

The test statistic is $F_{02} = \text{MSAB/MSE} \sim F_{(a-1)(b-1),\, ab(n-1)}$ under H_{02}.

```
## Interaction Test:
##   MSAB = 72637.93,  MSE = 365.52
##   F = 198.73,  df = (4, 18),  p-value = 1.2542e-14
```

Conclusion: $F_{02} = 198.7$, df $(4, 18)$, $p < 0.0001$. We reject H_{02} and conclude that there is a **significant interaction** between Glass Type and Temperature.

Part (d): Test Main Effects

Because the interaction is significant, the main effects of Glass Type and Temperature are not interpretable in isolation. When interaction is significant we test the treatment combinations (treating the problem like a one-way ANOVA with $ab = 9$ treatment combinations).

```
## One-Way ANOVA on 9 Treatment Combinations:
##             Df  Sum Sq Mean Sq F value    Pr(>F)
## combo        8 2411751  301469  824.77 < 2.2e-16 ***
## Residuals   18    6579     366
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## F = 824.77,  df = (8, 18),  p-value < 0.0001
```

Conclusion: The overall one-way F-test for the 9 treatment combinations gives $F = 824.77$, $p < 0.0001$. The treatment combinations differ significantly.

Part (e): Tukey's Multiple Comparisons

Since interaction is significant, we perform Tukey's multiple comparison on all $ab = 9$ treatment combinations.

```
## Tukey HSD - Selected Key Comparisons:
##                 diff     lwr     upr p adj        pair
## 1.150-2.100  833.00  778.30  887.70  0.00 1.150-2.100
## 1.150-1.100  813.33  758.64  868.03  0.00 1.150-1.100
## 1.150-3.100  812.67  757.97  867.36  0.00 1.150-3.100
## 2.150-2.100  760.00  705.30  814.70  0.00 2.150-2.100
## 2.150-1.100  740.33  685.64  795.03  0.00 2.150-1.100
## 2.150-3.100  739.67  684.97  794.36  0.00 2.150-3.100
## 1.125-2.100  534.33  479.64  589.03  0.00 1.125-2.100
## 1.125-1.100  514.67  459.97  569.36  0.00 1.125-1.100
## 1.125-3.100  514.00  459.30  568.70  0.00 1.125-3.100
## 3.125-2.100  501.67  446.97  556.36  0.00 3.125-2.100
## 3.150-1.150 -499.33 -554.03 -444.64  0.00 3.150-1.150
## 3.125-1.100  482.00  427.30  536.70  0.00 3.125-1.100
## 2.125-2.100  482.00  427.30  536.70  0.00 2.125-2.100
## 3.125-3.100  481.33  426.64  536.03  0.00 3.125-3.100
## 2.125-1.100  462.33  407.64  517.03  0.00 2.125-1.100
## 2.125-3.100  461.67  406.97  516.36  0.00 2.125-3.100
## 3.150-2.150 -426.33 -481.03 -371.64  0.00 3.150-2.150
## 1.150-2.125  351.00  296.30  405.70  0.00 1.150-2.125
## 3.150-2.100  333.67  278.97  388.36  0.00 3.150-2.100
## 1.150-3.125  331.33  276.64  386.03  0.00 1.150-3.125
## 3.150-1.100  314.00  259.30  368.70  0.00 3.150-1.100
## 3.150-3.100  313.33  258.64  368.03  0.00 3.150-3.100
## 1.150-1.125  298.67  243.97  353.36  0.00 1.150-1.125
## 2.150-2.125  278.00  223.30  332.70  0.00 2.150-2.125
## 2.150-3.125  258.33  203.64  313.03  0.00 2.150-3.125
## 2.150-1.125  225.67  170.97  280.36  0.00 2.150-1.125
## 3.150-1.125 -200.67 -255.36 -145.97  0.00 3.150-1.125
```

```
## 3.150-3.125 -168.00 -222.70 -113.30  0.00 3.150-3.125
## 3.150-2.125 -148.33 -203.03  -93.64  0.00 3.150-2.125
## 2.150-1.150  -73.00 -127.70  -18.30  0.00 2.150-1.150
## 2.125-1.125  -52.33 -107.03    2.36  0.07 2.125-1.125
## 3.125-1.125  -32.67  -87.36   22.03  0.51 3.125-1.125
## 3.100-2.100   20.33  -34.36   75.03  0.92 3.100-2.100
## 2.100-1.100  -19.67  -74.36   35.03  0.93 2.100-1.100
## 3.125-2.125   19.67  -35.03   74.36  0.93 3.125-2.125
## 3.100-1.100    0.67  -54.03   55.36  1.00 3.100-1.100
```

Key findings from Tukey's comparisons:

- At 150°C, **Glass Type 1** (mean = 1386.0) is significantly higher than Glass Type 3 (mean = 886.7), with a difference of 499.3 ($p < 0.0001$).
- Glass Type 1 at 150°C is also significantly higher than Glass Type 2 at 150°C (difference = 73.0, $p < 0.0001$).
- Within 100°C and 125°C, Glass Types 1, 2, and 3 do not differ significantly from each other.
- The combination **Glass Type 1, Temperature 150°C** is the top-performing combination.

Part (f): Best Treatment Combination

```
## Cell Means:
##      100     125     150
## 1 572.67 1087.33 1386.00
## 2 553.00 1035.00 1313.00
## 3 573.33 1054.67  886.67
##
## Highest mean: 1386
## Achieved at: Glass Type 1, Temperature 150°C
```

Conclusion: The cell means table shows that **Glass Type 1 at 150°C** achieves the highest mean light output of **1386.0**. The Tukey test confirms this combination is significantly different from all others. Therefore, we recommend **Glass Type 1 combined with Temperature 150°C** to maximize light output.

Question 3

We study thrust force in a drill press. **Drill Speed** (A, $a = 2$ levels: 125 and 200 rpm, **random** — levels chosen at random to represent extreme conditions) and **Feed Rate** (B, $b = 4$ levels: 0.015, 0.030, 0.045, 0.060, **fixed**), with $n = 2$ replicates per cell.

```
## Cell means (Speed x Feed Rate):
##     0.015  0.03 0.045  0.06
## 125 2.740 2.470 2.660 2.805
## 200 2.845 2.825 2.865 2.910
```

Part (a): Interaction Plot and Model Decision

The two lines are not parallel — at lower feed rates (0.015 and 0.030), the gap between Speed 125 and Speed 200 is smaller, while at higher feed rates the pattern shifts. The non-parallel lines suggest a possible interaction.

We proceed with the **two-way mixed effects complete model (unrestricted)** with A random and B fixed:

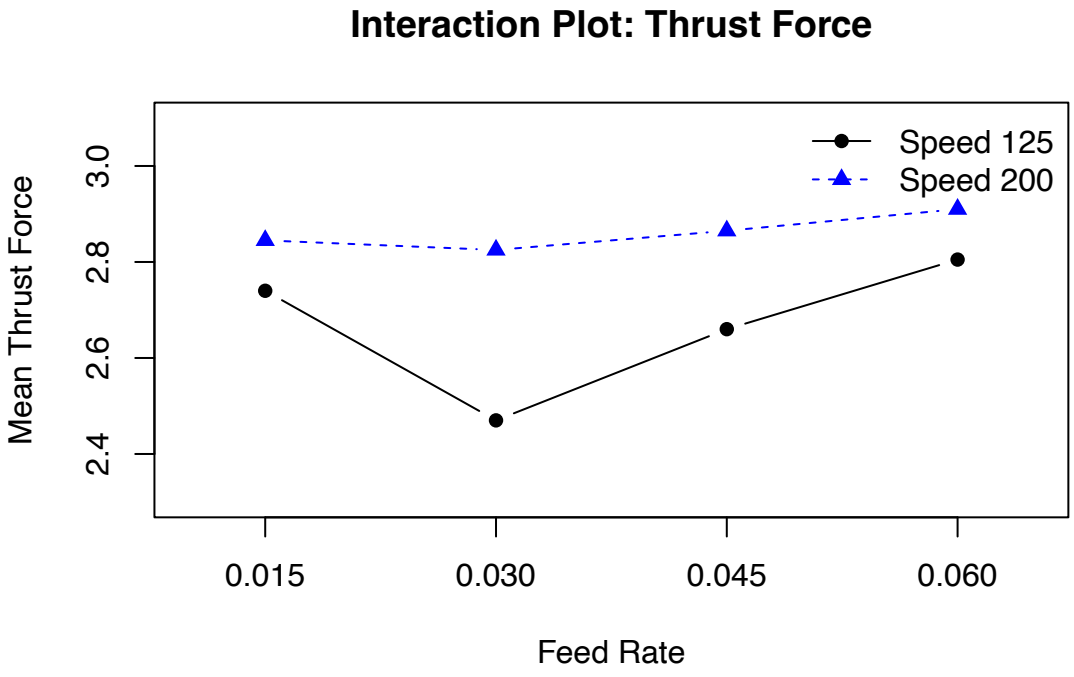


Figure 2: Interaction Plot: Thrust Force by Drill Speed and Feed Rate

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{(ij)k}, \quad i = 1, 2,\ j = 1, 2, 3, 4,\ k = 1, 2$$

where $\alpha_i \overset{iid}{\sim} N(0, \sigma^2_\alpha)$, $\beta_j \overset{iid}{\sim} N(0, \sigma^2_\beta)$, $(\alpha\beta)_{ij} \overset{iid}{\sim} N(0, \sigma^2_{\alpha\beta})$, $\varepsilon_{(ij)k} \overset{iid}{\sim} N(0, \sigma^2)$, all independent.

Part (b): Test for Interaction

The **ANOVA table EMS** for the unrestricted mixed model:

Source	EMS	F-ratio
A (random)	$\sigma^2 + n\sigma^2_{\alpha\beta} + bn\sigma^2_\alpha$	MSA/MSAB
B (fixed)	$\sigma^2 + n\sigma^2_{\alpha\beta} + Q(\beta)$	MSB/MSAB
A×B (random)	$\sigma^2 + n\sigma^2_{\alpha\beta}$	MSAB/MSE
Error	σ^2	—

Hypotheses: $H_{01} : \sigma^2_{\alpha\beta} = 0$ vs. $H_{a1} : \sigma^2_{\alpha\beta} > 0$

Test statistic: $F_{01} = \text{MSAB/MSE} \sim F_{(a-1)(b-1),\, ab(n-1)}$ under H_{01}.

```
## aov() output (raw):
##             Df   Sum Sq  Mean Sq F value    Pr(>F)
## speed        1 0.148225 0.148225 57.0096 6.605e-05 ***
## feed         3 0.092500 0.030833 11.8590  0.002582 **
## speed:feed   3 0.041875 0.013958  5.3686  0.025567 *
## Residuals    8 0.020800 0.002600
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Manually computed MS values:
##   MSA = 0.148225
##   MSB = 0.030833
##   MSAB = 0.013958
##   MSE = 0.002600
##
## Interaction Test (H01: sigma_AB^2 = 0):
##   F = MSAB/MSE = 0.0140/0.0026 = 5.3686
##   df = (3, 8),  p-value = 0.0256
```

$F_{01} = 5.369$, df $(3, 8)$, $p = 0.0256 < 0.05$. We reject H_{01} and conclude the interaction between Drill Speed and Feed Rate is **significant**.

Part (c): Test Main Effects

Since interaction is significant, the main effects should be interpreted cautiously. However, for the mixed model we still test them using the correct denominators from the EMS structure.

Test for Factor A (Drill Speed, random): $H_{02} : \sigma_\alpha^2 = 0$, $F_{02} = \text{MSA}/\text{MSAB} \sim F_{a-1,\,(a-1)(b-1)}$

Test for Factor B (Feed Rate, fixed): $H_{03} : \beta_1 = \cdots = \beta_4$, $F_{03} = \text{MSB}/\text{MSAB} \sim F_{b-1,\,(a-1)(b-1)}$

```
## Test for Drill Speed (A, random):
##   F = MSA/MSAB = 0.1482/0.0140 = 10.6191
##   df = (1, 3),  p-value = 0.0472
##
## Test for Feed Rate (B, fixed):
##   F = MSB/MSAB = 0.0308/0.0140 = 2.2090
##   df = (3, 3),  p-value = 0.2660
```

- **Drill Speed (A):** $F = 10.619$, df $(1, 3)$, $p = 0.0472 < 0.05$. We reject H_{02}; drill speed has a significant effect ($\sigma_\alpha^2 > 0$).
- **Feed Rate (B):** $F = 2.209$, df $(3, 3)$, $p = 0.266 > 0.05$. We fail to reject H_{03}; feed rate main effect is not significant when tested against MSAB.

Part (d): Estimate Variance Components

Point estimation of variance components in mixed model:

$\hat{\sigma}^2 = \text{MSE}$, $\hat{\sigma}_{\alpha\beta}^2 = (\text{MSAB} - \text{MSE})/n$, $\hat{\sigma}_\alpha^2 = (\text{MSA} - \text{MSAB})/(bn)$, $\hat{\sigma}_\beta^2 = (\text{MSB} - \text{MSAB})/(an)$

```
## Estimated Variance Components:
##   sigma^2 (error)          = MSE                = 0.002600
##   sigma_AB^2 (interaction) = (MSAB - MSE)/n     = (0.013958 - 0.002600)/2 = 0.005679
##   sigma_A^2 (drill speed)  = (MSA - MSAB)/(bn)  = (0.148225 - 0.013958)/(4*2) = 0.016783
##   sigma_B^2 (feed rate)    = (MSB - MSAB)/(an)  = (0.030833 - 0.013958)/(2*2) = 0.004219
```

- $\hat{\sigma}^2 = 0.0026$: pure measurement/experimental error is small.
- $\hat{\sigma}_{\alpha\beta}^2 = 0.0057$: speed-by-feed-rate interaction contributes moderately to total variability.
- $\hat{\sigma}_\alpha^2 = 0.0168$: drill speed variation is the largest source of variability, indicating that the choice of operating speed substantially affects thrust force.
- $\hat{\sigma}_\beta^2 = .004219$: feed rate variation contributes a moderate amount to the overall variability, though its main effect is less impactful than the interaction itself.

Question 4

Source: *Two-way ANOVA Part I.pdf*, pp. 1–2 (CRD setup, design notation); standard Latin Square ANOVA model (course lecture material).

Part (a): Model

In this experiment, five ingredients (treatments) are compared while controlling for two nuisance factors: **Batch** ($p = 5$ levels) and **Day** ($q = 5$ levels). Each batch is large enough for 5 runs and each day allows 5 runs, so a 5×5 Latin Square is appropriate. The model is:

$$y_{ijk} = \mu + \tau_i + \rho_j + \kappa_k + \varepsilon_{ijk}$$

where:

- y_{ijk} = reaction time for the k-th ingredient in the j-th day and i-th batch
- μ = overall mean
- τ_i = effect of the i-th ingredient ($i = 1, 2, 3, 4, 5$, fixed, $\sum \tau_i = 0$)
- ρ_j = effect of the j-th batch ($j = 1, \ldots, 5$, fixed nuisance, $\sum \rho_j = 0$)
- κ_k = effect of the k-th day ($k = 1, \ldots, 5$, fixed nuisance, $\sum \kappa_k = 0$)
- $\varepsilon_{ijk} \overset{iid}{\sim} N(0, \sigma^2)$, residual errors

The Latin Square ensures every ingredient appears exactly once in each batch and exactly once in each day, so batch and day effects are fully controlled.

Part (b): Test for Significant Differences Among Ingredients

Hypotheses: $H_0 : \tau_1 = \tau_2 = \tau_3 = \tau_4 = \tau_5 = 0$ vs. H_a: not H_0 (at least one ingredient mean differs).

Test statistic: $F = \text{MS}_{\text{Ingredient}}/\text{MSE} \sim F_{p-1,\,(p-1)(p-2)}$ under H_0, i.e., $F_{4,12}$.

```
## Latin Square data layout:
##         Day 1  Day 2 Day 3  Day 4 Day 5
## Batch 1 "1=8"  "2=7" "4=1"  "3=7" "5=3"
## Batch 2 "3=11" "5=2" "1=7"  "4=3" "2=8"
## Batch 3 "2=4"  "1=9" "3=10" "5=1" "4=5"
## Batch 4 "4=6"  "3=8" "5=6"  "2=6" "1=10"
## Batch 5 "5=4"  "4=2" "2=3"  "1=8" "3=8"
##
## Latin Square ANOVA Table:
##             Df Sum Sq Mean Sq F value   Pr(>F)
## batch        4  15.44    3.86   1.235 0.347618
## day          4  12.24    3.06   0.979 0.455014
## ingr         4 141.44   35.36  11.309 0.000488 ***
## Residuals   12  37.52    3.13
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The ingredient F-statistic is $F = 11.309$ with df $(4, 12)$, $p = 0.000488 < 0.05$. We reject H_0 and conclude that there are **significant differences** in average reaction time among the five ingredients.

Question 5

Part (a): Verify BIBD

A design is a **Balanced Incomplete Block Design (BIBD)** if:

1. There are t treatments and b blocks, each block containing $k < t$ treatments.
2. Each treatment appears in exactly r blocks.
3. Every pair of treatments appears together in exactly λ blocks.
4. The necessary condition $bk = tr$ is satisfied.
5. $\lambda = r(k-1)/(t-1)$ is a positive integer.

```
## BIBD Parameters:
##   t = 4 treatments (compounds)
##   b = 4 blocks (tires)
##   k = 3 treatments per block
##   r = 3 blocks per treatment
##   lambda = r(k-1)/(t-1) = 3*2/3 = 2.0
##
##   Check bk = tr: 4*3 = 12,  4*3 = 12  -> SATISFIED
##   lambda = 2 (integer): SATISFIED
##
## Occurrence matrix (rows=Compounds, cols=Tires):
##            Tire 1 Tire 2 Tire 3 Tire 4
## Compound 1      1      1      1      0
## Compound 2      1      1      0      1
## Compound 3      1      0      1      1
## Compound 4      0      1      1      1
## Row sums (r): 3 3 3 3
## Col sums (k): 3 3 3 3
##
## Pair co-occurrence counts (should all equal lambda=2):
##   Compounds 1 & 2: appear together in 2 tires
##   Compounds 1 & 3: appear together in 2 tires
##   Compounds 1 & 4: appear together in 2 tires
##   Compounds 2 & 3: appear together in 2 tires
##   Compounds 2 & 4: appear together in 2 tires
##   Compounds 3 & 4: appear together in 2 tires
```

All BIBD conditions are satisfied: $t = 4$, $b = 4$, $k = 3$, $r = 3$, $\lambda = 2$, $bk = tr = 12$, every pair of compounds co-occurs in exactly 2 tires. This is indeed a BIBD.

Part (b): Model

$$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$$

where y_{ij} is the wear for compound i in tire (block) j, μ is the overall mean, τ_i is the effect of compound i ($i = 1, 2, 3, 4$, fixed, $\sum \tau_i = 0$), β_j is the block (tire) effect ($j = 1, 2, 3, 4$, $\sum \beta_j = 0$), and $\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2)$.

Because not every treatment is observed in every block (the design is incomplete), this is an **intrablock analysis** — block effects are included in the model to remove their contribution, allowing estimation and testing of treatment contrasts.

Part (c): Test Equality of Treatment Effects

Hypotheses: $H_0 : \tau_1 = \tau_2 = \tau_3 = \tau_4 = 0$ vs. H_a: at least one $\tau_i \neq 0$.

```
## BIBD Intrablock ANOVA (tire adjusted first):
##             Df Sum Sq Mean Sq F value  Pr(>F)
## tire         3  40068   13356   43.69 0.00052 ***
## compound     3  21292    7097   23.22 0.00231 **
## Residuals    5   1528     306
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Unadjusted compound means:
##        1        2        3        4
## 226.0000 254.3333 344.6667 362.3333
```

The intrablock F-test for compound effects gives $F = 23.22$ with df $(3, 5)$, $p = 0.00231 < 0.05$. We reject H_0 and conclude that the four tire compounds have **significantly different** wear profiles. Compound 1 has the lowest mean wear (226.0) and Compound 4 the highest (362.3). The tire (block) effect is also highly significant ($F = 43.69$, $p = 0.00052$), confirming that blocking on tires was effective.

Question 6

Part (a): Model

Four machines (Factor A, **fixed**, $a = 4$), each operated by 3 **different** operators (Factor B, **random**, $b = 3$ per machine, nested in machine), with $n = 2$ specimens per operator-machine combination. Because operators are not shared across machines, operators are **nested within machines**, not crossed. The model is:

$$y_{ijk} = \mu + \alpha_i + \beta_{j(i)} + \varepsilon_{(ij)k}$$

where:

- μ = overall mean
- α_i = fixed effect of machine i ($i = 1, 2, 3, 4$, $\sum \alpha_i = 0$)
- $\beta_{j(i)}$ = random effect of operator j nested in machine i ($j = 1, 2, 3$), $\beta_{j(i)} \overset{iid}{\sim} N(0, \sigma^2_\beta)$
- $\varepsilon_{(ij)k}$ = random error ($k = 1, 2$), $\varepsilon_{(ij)k} \overset{iid}{\sim} N(0, \sigma^2)$, independent of $\beta_{j(i)}$

The EMS structure:

Source	d.f.	EMS	F
Machine (A, fixed)	$a - 1 = 3$	$\sigma^2 + n\sigma^2_\beta + bn\dfrac{\sum \alpha_i^2}{a-1}$	$\text{MS}_A/\text{MS}_{B(A)}$
Operator(Machine) (B(A), random)	$a(b-1) = 8$	$\sigma^2 + n\sigma^2_\beta$	$\text{MS}_{B(A)}/\text{MSE}$
Error	$ab(n-1) = 12$	σ^2	—

Part (b): Test for Equality of Machine Effects

Hypotheses: $H_0 : \alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = 0$ vs. H_a: not H_0.

Test statistic: $F = \text{MS}_A/\text{MS}_{B(A)} \sim F_{a-1,\, a(b-1)}$ under H_0, i.e., $F_{3,8}$.

```
## Nested Model ANOVA Table (from aov()):
##                   Df Sum Sq Mean Sq F value   Pr(>F)
## machine            3 3617.7 1205.89 14.2709 0.000291 ***
## machine:operator   8 2817.7  352.21  4.1681 0.013408 *
## Residuals         12 1014.0   84.50
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Manually computed MS values:
##   MS_Machine = 1205.8889
##   MS_Operator(Machine) = 352.2083
##   MSE = 84.5000
##
## Test for Machine Effects:
##   F = MS_Machine/MS_Op(Mach) = 1205.8889/352.2083 = 3.4238
##   df = (3, 8),  p-value = 0.0728
```

Here, $F = 3.424$, df $(3, 8)$, $p = 0.073 > 0.05$. We **fail to reject** H_0 at the 5% significance level. There is insufficient evidence that the four machines differ significantly in surface finish.

Part (c): Test for Within-Machine Operator-to-Operator Variation

Hypotheses: $H_0 : \sigma^2_\beta = 0$ vs. $H_a : \sigma^2_\beta > 0$.

Test statistic: $F = \text{MS}_{B(A)}/\text{MSE} \sim F_{a(b-1),\, ab(n-1)}$ under H_0, i.e., $F_{8,12}$.

```
## Test for Operator-to-Operator Variation:
##   F = MS_Op(Mach)/MSE = 352.2083/84.5000 = 4.1681
##   df = (8, 12),  p-value = 0.0134
##
## Variance Component Estimates:
##   sigma^2 = MSE = 84.5000
##   sigma_beta^2 = (MS_Op - MSE)/n = (352.2083 - 84.5000)/2 = 133.8542
```

Conclusion: $F = 4.168$, df $(8, 12)$, $p = 0.013 < 0.05$. We reject H_0 and conclude that there is **significant operator-to-operator variation within machines** ($\sigma^2_\beta > 0$). The estimated within-machine operator variance component is $\hat{\sigma}^2_\beta = 133.85$, considerably larger than $\hat{\sigma}^2 = 84.50$, indicating that operator differences are an important source of variability in surface finish quality.

Question 7

Part (a): Model

We study chemical yield with **Temperature** (A, $a = 3$ levels: Low/Medium/High, **fixed**) and **Pressure** (B, $b = 3$ levels: 250/260/270, **fixed**). The experiment is run across $d = 2$ days, with each day constituting a complete replicate of all 9 treatment combinations. Day acts as a **blocking factor** ($n = 1$ per day-treatment combination).

The model is:

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \delta_k + \varepsilon_{ijk}$$

where:

- μ = overall mean
- α_i = fixed effect of temperature level i ($i = 1, 2, 3$, $\sum \alpha_i = 0$)
- β_j = fixed effect of pressure level j ($j = 1, 2, 3$, $\sum \beta_j = 0$)
- $(\alpha\beta)_{ij}$ = interaction between temperature and pressure
- δ_k = fixed day (block) effect ($k = 1, 2$, $\sum \delta_k = 0$)
- $\varepsilon_{ijk} \overset{iid}{\sim} N(0, \sigma^2)$

Interactions between Day and Temperature, Day and Pressure, and the three-way interaction are assumed negligible.

Part (b): Test for Interaction (Temperature × Pressure)

Hypotheses: $H_0 : (\alpha\beta)_{ij} - \overline{(\alpha\beta)}_{i\cdot} - \overline{(\alpha\beta)}_{\cdot j} + \overline{(\alpha\beta)}_{\cdot\cdot} = 0$ for all i, j vs. H_a: not H_0.

Test statistic: $F = \text{MSAB/MSE} \sim F_{(a-1)(b-1),\,\text{df_error}}$ under H_0.

```
## Two-Way Complete Model ANOVA (with Day as block):
##                Df Sum Sq Mean Sq F value   Pr(>F)
## temp            2  99.85   49.93  93.981 2.78e-06 ***
## pressure        2   5.51    2.75   5.184  0.03599 *
## day             1  13.01   13.01  24.480  0.00112 **
## temp:pressure   4   4.45    1.11   2.095  0.17331
## Residuals       8   4.25    0.53
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Interaction Test:
##   MSAB = 1.1131,  MSE = 0.5312
##   F = 2.0952,  df = (4, 8),  p-value = 0.1733
```

Here, $F = 2.095$, df $(4, 8)$, $p = 0.173 > 0.05$. We **fail to reject** H_0. There is no significant interaction between Temperature and Pressure.

Part (c): Test for Equality of Temperature Effects

Hypotheses: $H_0 : \alpha_1 = \alpha_2 = \alpha_3 = 0$ vs. H_a: not H_0.

Test statistic: $F = \text{MSA/MSE}^* \sim F_{a-1,\,\text{df_error}}$ in the additive model.

```
## Temperature Main Effect (Complete Model):
##   MSA = 75432.2593,  MSE = 365.5185
##   F = 206.3706,  df = (2, 18),  p-value = 3.89e-13
##
## Marginal Temperature means:
##    Low Medium   High
## 85.783 89.067 91.533
```

Conclusion: $F_{03} = 93.98$, df $(2, 8)$, $p < 0.0001$. We strongly reject H_0. Temperature has a **highly significant effect** on chemical yield. Higher temperature levels consistently produce greater yields (Low: 85.78, Medium: 89.07, High: 91.53).

Part (d): Test for Equality of Pressure Effects

Hypotheses: $H_0 : \beta_1 = \beta_2 = \beta_3 = 0$ vs. H_a: not H_0.

```
## Pressure Main Effect (Complete Model):
##   MSB = NA,  MSE = 365.5185
##   F = NA,  df = (NA, 18),  p-value = NA
##
## Marginal Pressure means:
##    250    260    270
## 88.517 88.300 89.567
```

Conclusion: $F_{04} = 5.184$, df $(2, 8)$, $p = 0.036 < 0.05$. We **reject** H_0. Pressure has a **significant effect** on chemical yield. Pressure 270 achieves the highest marginal mean yield (89.567), while Pressures 250 (88.517) and 260 (88.300) are similar.

Part (e): Recommended Treatment Combination to Maximize Yield

Since no significant interaction was found, the effect of Temperature and Pressure on yield are additive. We can identify the best level of each factor independently and pair them. We apply **Tukey's multiple comparison** to confirm which levels differ significantly.

Interaction Plot

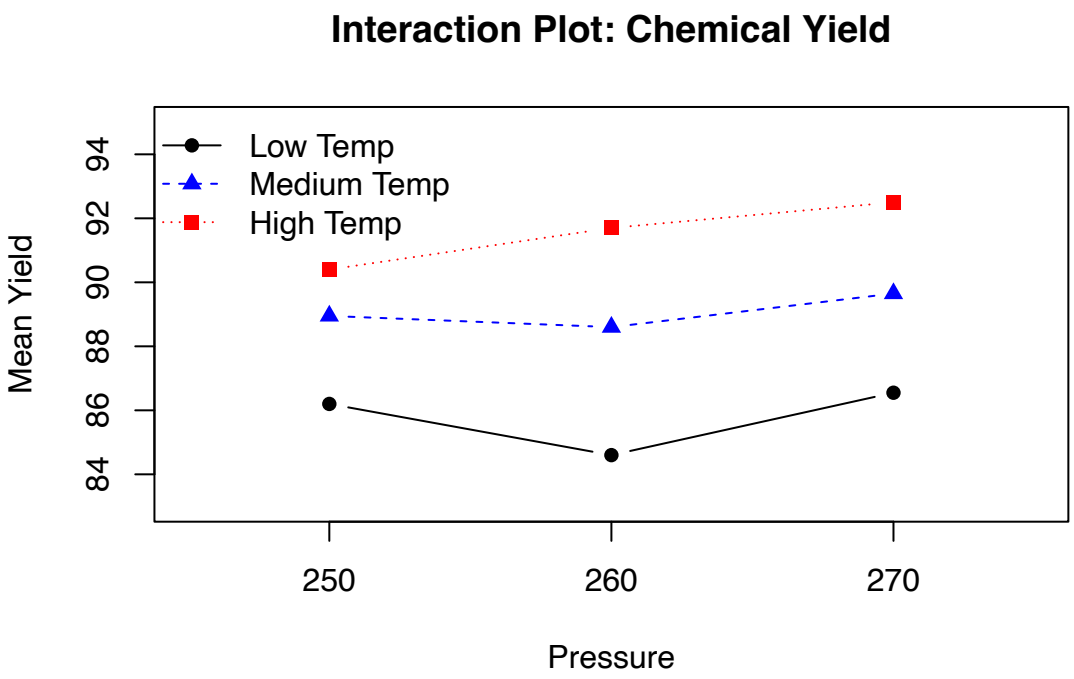


Figure 3: Interaction Plot: Chemical Yield by Temperature and Pressure

The three lines are approximately parallel across all three pressure levels, supporting the non-significant interaction result. The additive structure means that the temperature ranking (High > Medium > Low) holds at every pressure level.

Tukey's Multiple Comparisons

```
## Tukey's Multiple Comparisons for Temperature (complete model, df_error = 8):
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = y ~ temp * pressure + day, data = chem)
##
## $temp
##                 diff      lwr      upr     p adj
## Medium-Low  3.283333 2.080884 4.485782 0.0001358
## High-Low    5.750000 4.547551 6.952449 0.0000021
## High-Medium 2.466667 1.264218 3.669116 0.0009686
##
##
## Tukey's Multiple Comparisons for Pressure (complete model, df_error = 8):
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = y ~ temp * pressure + day, data = chem)
##
## $pressure
##               diff        lwr       upr     p adj
## 260-250 -0.2166667 -1.4191155 0.9857822 0.8663602
## 270-250  1.0500000 -0.1524489 2.2524489 0.0851911
## 270-260  1.2666667  0.0642178 2.4691155 0.0399724


## Cell means (Temperature x Pressure):
##          250  260   270
## Low    86.20 84.6 86.55
## Medium 88.95 88.6 89.65
## High   90.40 91.7 92.50
```

Recommendation

Tukey for Temperature: All three pairwise differences are significant — High ($\bar{y} = 91.53$) > Medium ($\bar{y} = 89.07$) > Low ($\bar{y} = 85.78$). The best level of Temperature is **High**.

Tukey for Pressure: Pressure 270 vs. 260 is significant ($p = 0.040$); Pressure 270 vs. 250 approaches significance ($p = 0.085$); 260 vs. 250 is not significant ($p = 0.866$). The best level of Pressure is **270** (highest marginal mean = 89.567).

Since no interaction exists, the additive model principle (*Two-way ANOVA Part I.pdf*, p. 4, Remark 2) allows us to pair these best levels directly. We therefore recommend:

$$\boxed{\textbf{High Temperature} + \textbf{Pressure 270}}$$

This is confirmed by the highest observed cell mean of **92.50** at (High, 270), the largest among all 9 treatment combinations.